

The Water Cycle

Water moves between the oceans, the air and the land in a continuous loop. The sun warms the surface of the sea and the water evaporates into the air as vapour.

High in the sky the vapour cools and condenses into tiny droplets, which gather as clouds. When the droplets grow heavy they fall as rain, snow or hail, and the water collects again in rivers, lakes and the ground.

- evaporation
- condensation
- precipitation
- collection

Energy and Motion

Three formulas that every student meets early on.

Kinetic energy of a moving body:

$$E = \frac{1}{2} mv^2$$

The sides of a right triangle:

$$a^2 + b^2 = c^2$$

The roots of a quadratic equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Notes on Memory

Things are remembered better when they are recalled than when they are reread. A short test after reading does more for memory than a second reading of the same page.

Spacing helps too: seeing a fact again just before it would be forgotten makes it last longer.

- recall beats rereading
- space the repetitions
- sleep after learning